Tom Milner

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PERSONAL PROFILE

I am experienced in various programming languages, embedded electronics, mechanical design and 3D printing, and graduated at the top of the cohort at master's level mechatronics and undergraduate computer engineering. I have experience in team-based technical roles, and I am friendly, hard-working, and punctual. I have won scholarships and awards for both academic and professional achievements, and I pride myself on my communication, willingness to learn new skills, and self-motivation.

RELEVANT WORK EXPERIENCE

BACKEND ENGINEER, STINT**,** January 2026 – Present

- Write **Go** and **Python** for various backend services in an **Agile** team.
- Write **Flutter** for frontend apps and websites.
- Configure **gRPC** communications between microservices.
- Configure **Gitlab CI/CD pipelines** for automated deployment.
- Manage **PostreSQL** databases.
- Utilised **Agentic Workflows** using Claude Code / Cursor.

ROBOTICS SOFTWARE ENGINEER, OC ROBOTICS (GE AEROSPACE)**,** July 2022 – July 2023

- Contributed embedded **C, C++, Python**, and **JavaScript** to the core codebases of "snake arm" robots.
- Wrote deployment tooling for Linux (**bash**) and Windows (**PowerShell**) systems.
- Configured **Nvidia Jetson TX2**s to run custom Linux images.
- Utilised **UART, I2C, UART** to connect to peripherals (hexapod, motor drivers).
- Configured **Ethernet** network switch for internal robot network.
- Scripted "**Universal Robotics**" robots to automate camera calibration processes.
- Earned 5 awards over the year for various software contributions to the company.

FULLSTACK WEB DEVELOPER, APOLLO SOFTWARE**,** March 2019 - September 2020

FEATURED PROJECTS

IN-PIPE INSPECTION ROBOT (MSC THESIS, 86%), FUSION 360, PYTHON, DFM

Designed and prototyped an in-pipe inspection robot designed to traverse underground pipes of 50mm diameter capable of a max speed of 234 mm/s, making it the fastest of its class in the literature. Designed a process for generating and optimising cylindrical cams with integrated ball bearings, allowing high-velocity mm-scale linear actuators. The robot was designed with Fusion 360 and printed using a Bambu Lab X1C printer. The project taught me parametric design in CAD, as well as DFMA, and was awarded 86%.

"KLIP", A HOUSEPLANT WALL HANGER, FUSION 360, PRODUCT DESIGN, DFM

Designed and built a homeware product that allows a user to invisibly hang a plant pot on the wall with temporary fasteners. Designed in Fusion360 and prototyped using 3D printing. Performed extensive failure mode analysis and load testing on the product, before pitching it to investors, where I was awarded First Place in the SPARK Business Plan Competition and awarded £2500.

"VINYL", A TACTILE MUSIC STREAMER, PRODUCT DESIGN, ESP32, PLATFORMIO

Designed and built an audio streaming device to enable interacting with digital audio streaming services in a tactile way. The experience is designed to mimic that of a vinyl turntable. Different "cards" each contain an RFID sticker encoding the Spotify ID of a piece of content (playlist, album, artist, song etc). When a card is placed on the player, an ESP32 in the player uses an RFID reader to read the Spotify ID and plays the content through integrated speaker drivers, or a Spotify Connect device on the network. I designed and printed an enclosure, and pitched the product to a city-wide enterprise competition, where I was awarded £2000 to develop the product further.

FPGA OSCILLOSCOPE, VERILOG

Built a digital oscilloscope implemented on the DE1-SoC FPGA platform featuring channel selection, adjustable trigger levels, x-axis scaling, frequency detection, and a real time waveform display on an LCD. I was responsible for integrating the on-board ADC chip of the DE1-SoC. Quartus Prime was used to develop, simulate and test the ADC integration. The project was awarded 84%.

INTERNET RADIO STREAMER (NTS.LIVE), FUSION 360, RASPBERRY PI, PYTHON

Building a device to stream internet radio via m3u / HLS streams. The system runs on a Raspberry PI and integrates a "DigiAmp+" DAC and amplifier to power full range speaker drivers. Designed an enclosure using Fusion 360, 3D printed on a Prusa MK3S+. Currently iterating on the design to improve audio quality by learning about building speaker enclosures.

AUDIO EFFECTS UNIT, MATLAB, C++, JUCE

Using Digital Signal Processing (DSP) techniques, I analysed and implemented various common audio effects: echo, distortion, fuzz, reverb, tremolo, 'auto-wah', and a vocoder. Each audio effect was prototyped in MATLAB, before being implemented in C++ as a VST plugin (JUCE) for real-world usage with an electric guitar. I learnt about feedforward and feedback comb filters, impulse responses, frequency responses, allpass filters, pole-zero plots, convolutional and algorithmic reverb implementations, and more.

SPOTIFY-SYNCHRONISED LED STRIPS, GO, C++, ESP32 (ARDUINO)

Built a system to synchronise the intensity and colour of multiple LED strips to a user's currently playing Spotify content. A single Go "Gateway" server runs on a PC. It connects to the Spotify Web API, authenticates using OAuth2, and downloads the album cover art and timestamped "beats" and "bars" of the currently playing track. The Go server computes the dominant colour of the album art, before broadcasting it (and the beat information) via MQTT to C++ "Edge" applications running on ESP32s. Each ESP32 uses the beat and colour information to pulse an LED strip in time with the music, setting the lights to the same colour as the album art.

MONOPHONIC SYNTHESIZER, C, ARM CORTEX A9

Implemented a monophonic audio synthesiser and sequencer on a DE1-SoC development board in C. Developed a UI engine to handle interaction through buttons and an LCD screen, note programming through a sequencer interface, waveform selection, and audio output drivers via I2S to a line out. I built a modular component-based user interface library to render buttons, labels, pages, page navigation controls, and sequencer controls, and modified the vendor display driver to improve display rendering times by 17%. I also utilised an oscilloscope logic analyser to debug various audio related bugs. The project was awarded 86%.

EDUCATION

MECHATRONICS AND ROBOTICS MSC(ENG), UNIVERSITY OF LEEDS, 2024 – 2025

- Academic Scholarship.
- **Distinction** (80%), coming **second in the cohort**.
- Top performing modules: Engineering Computational Methods (91%), Embedded Microprocessor System Design (86%), FPGA System Design (85%), AI and Biomechatronics (80%).

ELECTRONICS AND COMPUTER ENGINEERING BENG, UNIVERSITY OF LEEDS, 2020 - 2024

- Academic Scholarship.
- **First Class** (79%), coming **top of the cohort**.
- Top performing modules: FPGA Design (79%), Embedded Systems (79%), Digital Comms (92%), Compiler Design (97%), User Interfaces (88%), Circuit Design (79%), C Programming (99%).

A-LEVELS, GCSES CHURCHER'S COLLEGE, 2018 - 2020

- A-Levels: Mathematics (A*), Computer Science (A*), Physics (A).
- GCSEs: Three 9s, two 8s, five 7s.

INTERESTS

MUSIC PRODUCTION / MUSIC TECH

- Avid music fan - I love going to gigs, and also producing my own music using Ableton and a collection of mixers/synths/drum machines/guitar/pedals.